

MOHAMED RIZWAN AMEER JOHN

Embedded Software Engineer — Firmware, Real-Time Control & Hardware/Software Integration

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PROFESSIONAL SUMMARY

Embedded Software Engineer (MS Computer Science, May 2026; BE Electrical & Electronics Engineering) who designs, builds and debugs real-time embedded systems on physical hardware: bare-metal C/C++ on ARM Cortex-M and embedded Linux, FreeRTOS, device drivers and CAN/I2C/SPI/UART/BLE interfaces. Works requirements-first, with traceability, host-run unit tests in CI, MIL/SIL/HIL validation and bench root-cause analysis. Also brings MATLAB/Simulink model-based design, Python test automation and three filed patents.

TECHNICAL SKILLS

Languages: C, C++ (object-oriented design), Embedded C, Python (scripting, test automation, NumPy, OpenCV), MATLAB/Simulink (Stateflow, automatic code generation), Java, Bash (basic), SQL (basics)

Embedded Software & Firmware: bare-metal C/C++ on ARM Cortex-M (STM32 Cortex-M4, RP2040 Cortex-M0+) and Arduino AVR/ARM; FreeRTOS; embedded Linux on ARM Cortex-A72 (Raspberry Pi 4B); hardware abstraction layers (HAL); state machines; interrupt-driven I/O; real-time and deterministic control loops; device drivers; board and sensor bring-up; register-level debugging; GPIO/ADC/PWM/timers; GNU toolchain (GCC/AVR); SWD/JTAG via ST-Link and bootloaders (in progress)

Test, Verification & Debugging: software requirements and traceability matrices; test-case design with pass/fail criteria; unit testing (Unity, CMake/CTest); GitHub Actions CI with cppcheck static analysis and ARM cross-compile; Model-, Software- and Hardware-in-the-Loop (MIL/SIL/HIL); system integration testing; issue reproduction, defect logging and retest; root cause analysis; embedded debuggers and simulators; oscilloscope and DMM bench debugging; logic analyzer (in progress); ISO 26262, DFMEA and HARA (learning)

Protocols & Networking: CAN Bus, SAE J1939 (fundamentals), J1939-style PDU design with rolling counter and checksum, I2C, SPI, UART, BLE, TCP/IP; LIN and Automotive Ethernet (familiar); Wi-Fi and LTE (awareness); CANalyzer/CANoe (awareness)

Model-Based Design & Controls: MATLAB/Simulink and Stateflow with autocode, plant modeling, simulation test campaigns; closed-loop feedback and PID control, Adaptive Sliding Mode Control, Lyapunov stability, Bode and phase-margin analysis; Q-Learning navigation; D-H forward/inverse kinematics

Hardware, Power & Lab: reading schematics, datasheets and timing diagrams; DC-DC buck and boost converter design, power distribution, Li-Po systems; H-bridge and PWM servo motor drives; solenoid valves and pneumatic circuits; MPU6050 IMU, ultrasonic, flex and CSI camera sensors; wiring harness fabrication, soldering and crimping, PCB design; sensor calibration and signal-level troubleshooting

Tools & Practices: Git/GitHub, Linux, SOFA FEA, OpenCV, MediaPipe, TinyML, AutoCAD (basic); dSPACE, NI VeriStand (familiar); SolidWorks, Altium (learning); design documentation, code review, AI coding assistants

EDUCATION

MS Computer Science

2024 – May 2026

Governors State University — University Park, IL

BE Electrical and Electronics Engineering

2019 – May 2023

KCG College of Technology — Chennai, India

PROFESSIONAL EXPERIENCE

Build Student Consultant — Automotive Embedded Control Systems Analysis

Mar 2025 – Apr 2025

The Build Fellowship (Tesla FCW/AEB Study) | Remote

- Analyzed safety-critical embedded control system behavior (Tesla Model 3 FCW/AEB) against real-world crash-test data across 25–75 mph scenarios; quantified time-to-collision degradation from 2.1s at 25 mph to 1.3s at 75 mph and a 7.7% reduction in avoidance rate under partial-offset conditions, using the data to root-cause controller failure modes.
- Proposed perception and sensor-fusion enhancements for partial-object recognition to reduce false negatives; translated field validation findings into documented software improvement recommendations for cross-functional review.

PROJECT EXPERIENCE

CAN Vehicle Network Gear Controller — Embedded C, FreeRTOS, Two-Node CAN Bus

Sep 2026 – Present

Personal Project | Embedded C, STM32 (ARM Cortex-M4), FreeRTOS, CAN Bus, Unity, GitHub Actions

- Building a two-node transmission gear controller on STM32 with FreeRTOS: concurrent tasks for control, CAN transmit/receive and fault monitoring; a gear state machine with safety interlocks (brake-to-shift, speed-limited reverse and park); a shift scheduler with 3.0 km/h hysteresis; and J1939-style 8-byte CAN frames with rolling counter and checksum for message integrity and bus-loss detection.
- Architected the control logic as hardware-independent modules behind a HAL so it runs on a host machine: 33 Unity unit tests execute in CI on every push alongside cppcheck static analysis and an ARM cross-compile; authored 23 numbered requirements traced to their verifying tests.

Autonomous Mobile Robot — Embedded C++ Control, MIL to HIL Validation

May 2024

University of New Haven | C++, Arduino (AVR/ARM), L293D H-Bridge Drives, Ultrasonic Sensing, Q-Learning, HIL

- Built the robot end to end by hand (wiring harness, HC-SR04 ultrasonic sensing, L293D H-bridge PWM motor drives), then wrote embedded C++ motion-control and Q-Learning navigation software in object-oriented modules.
- Validated in MATLAB/Python Model-in-the-Loop simulation across 100+ episodes, then Hardware-in-the-Loop, then on the robot; reproduced and root-caused sensor calibration, PWM timing and control-loop defects across 100+ hardware trials until behavior was stable and repeatable.

EV Powertrain Boost Converter — Model-Based Design & Nonlinear Control (FCHEV)

May 2023

Research Project, University of New Haven | MATLAB/Simulink, Stateflow, Autocode

- Designed a nonlinear DC-DC boost converter for a fuel-cell hybrid electric vehicle powertrain ($L = 383 \mu\text{H}$, $C = 257 \mu\text{F}$, 12–18 V to 36 V, 100 W, 30 kHz) and its Adaptive Sliding Mode Controller; derived a Lyapunov stability proof for zero steady-state error and implemented supervisory logic as Stateflow state machines with automatic code generation.
- Ran a verification campaign across the full 1 W–100 W load range: 45 deg phase margin and 10 kHz gain crossover confirmed by Bode analysis, current ripple <5%, voltage ripple <0.5%. Co-authored the resulting research paper.

Smart Sign Language Eyewear — Wearable Device, Real-Time Video on Embedded Linux [Patent Filed]

Jan 2023

KCG College of Technology | Raspberry Pi 4B (ARM Cortex-A72), Embedded Linux, Python, OpenCV, MediaPipe

- Designed and built a battery-powered wearable that interprets Indian Sign Language in real time: 5 MP CSI camera video → OpenCV preprocessing → MediaPipe 21-point hand landmarks → CNN classifier → text-to-speech, fully on-device on embedded Linux with no cloud dependency.
- Designed the device hardware for a head-worn form factor (Li-Po 3.7 V/3250 mAh pack, LM2596 buck regulation, compact packaging). First real-time system for this task. Indian Patent Application filed Feb 10, 2023.

Smart Knee Actuator — Wearable Rehabilitation Device, Bare-Metal Firmware [Patent No. 202341027059]

Aug 2022

Smart India Hackathon | Embedded C, RP2040 (ARM Cortex-M0+), I2C, ADC, GPIO, BLE, TinyML, Pneumatics

- Wrote bare-metal C firmware on an RP2040 that samples an MPU6050 IMU over I2C and flex sensors via ADC, runs an on-device TinyML gait-phase classifier, and drives solenoid-operated 5/2 directional control valves through GPIO with deterministic timing to actuate a pneumatic double-acting cylinder, with BLE telemetry for live monitoring.
- Designed the electrical and fluid-power system (LM2596 buck rail, Li-Po pack, wiring harness, 10 bar supply, FRL unit, flow control), then commissioned it on the bench with oscilloscope and DMM, root-causing actuation timing, pressure regulation and sensor calibration faults. FEA validated to 500 N.

Soft Robotic Glove — Post-Stroke Hand Rehabilitation (Team Lead, 12 Members)

Jan 2022

KCG College of Technology | D-H Kinematics, Servo Drives, Soft Pneumatic Actuators, SOFA FEA

- Led a 12-member interdisciplinary team building a soft robotic glove for post-stroke rehabilitation; ran design reviews to resolve conflicting electrical, mechanical and software requirements.
- Modeled 5-finger multi-DOF kinematic chains with Denavit–Hartenberg parameter tables and designed servo and soft-pneumatic actuation for human-safe force control, validating the structure through FEA in SOFA before build.

PATENTS & ACHIEVEMENTS

- **Smart Knee Actuator** — Indian Patent No. 202341027059, published Oct 2024.
- **Sign Language Interpreting Eyewear** — Indian Patent Application filed Feb 10, 2023.
- **Wireless Sensor Network Management Web Platform** — Indian Patent No. 202341008938, Feb 2023.
- **IEEE YESIST12 International Finalist** — IEEE Egypt, Feb 2023.
- **IIT PALS Winner** — IIT Madras, Oct 2022.
- **Team Leadership** — led a 12-member interdisciplinary engineering team through a competitive two-round selection.